

Identifying Instructional Design Debt in Online Technical Learning Resources: A Systematic Literature Review

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1 Reproducibility and Audit Trail

To make the review reproducible by researchers without programming expertise, the companion materials are organized as a manual audit package rather than as an executable software requirement. The package preserves the original database exports, the exact search strings and filters, and a versioned workbook that records every decision from identification to synthesis. Each study receives a stable identifier that is used consistently across deduplication, screening, quality appraisal, extraction, and the evidence matrix.

The historical search can be reproduced from the preserved exports and search log, whereas a later rerun of the same query may retrieve a different number of records because bibliographic databases are continuously updated. Therefore, the search date, database interface, query, filters, result count, export format, and file name are recorded for every source. Screening and extraction decisions are made by human reviewers. Any automated or semantic support is identified as assistance only and is not treated as a final inclusion, exclusion, coding, or synthesis decision.

For non-technical use, the complete audit trail can be maintained in a single spreadsheet workbook with the following sheets: README, SEARCH_LOG, MASTER_RECORDS, DUPLICATES, ABSTRACT_SCREENING, FULLTEXT_SCREENING, CONSENSUS, QUALITY_APPRAISAL, EXTRACTION, EVIDENCE_MATRIX, PRISMA_COUNTS, and CHANGE_LOG. The PRISMA diagram and all reported totals should be derived from the final values in PRISMA_COUNTS, avoiding independent manual transcription of numbers in figures and tables.

Minimum reproducibility checks. Before publication, the research team should verify that: (i) every database has an exact query and execution date; (ii) raw exports remain unchanged; (iii) every excluded full-text study has one recorded

primary reason; (iv) disagreements have a documented consensus decision; (v) every extracted finding has a quotation and location; (vi) inferred instructional-design-debt evidence is explicitly marked; (vii) all PRISMA totals reconcile arithmetically; and (viii) the archived package corresponds to the manuscript version submitted.

2 Review Protocol and Reproducible Search Strategy

This section documents the review protocol, database-specific search strategies, screening rules, and quality-appraisal criteria in a form that can be followed without programming. The review methodology follows Kitchenham and Charters, whereas PRISMA 2020 is used as the reporting framework. The historical search is reproduced from the preserved database exports and search log; a later rerun may retrieve a different number of records because bibliographic databases are continuously updated.

2.1 Methodological protocol

2.2 Goal–Question–Metric formulation

The protocol addresses the following research questions:

- RQ1** Which acceptance criteria are used to evaluate online technical learning resources, and how are they aligned with taxonomies of knowledge use?
- RQ2** Which methods and indicators are used to identify or measure instructional design debt?

2.3 Database selection

2.4 Database-specific search strings

The following strings should be copied exactly into the corresponding advanced search interface. For every execution, the search date, interface, filters, result count, export format, and exported file name should be recorded in the search log.

Scopus.

```
TITLE-ABS-KEY(("instructional design" OR "technology acceptance" OR
"learning design" OR "design debt" OR "knowledge taxonomy" OR "Bloom*
taxonomy") W/15 ("online learning" OR "e-learning" OR MOOC OR "online
course*"))
AND TITLE-ABS-KEY("engineering education" OR "technical education" OR
"computing education")
AND PUBYEAR > 2004 AND PUBYEAR < 2026
AND (LANGUAGE(english) OR LANGUAGE(spanish))
AND (DOCTYPE(ar) OR DOCTYPE(cp) OR DOCTYPE(re))
AND (SUBJAREA(comp) OR SUBJAREA(engi) OR SUBJAREA(soci))
```

Web of Science.

```
TS=((("online learning" OR "e-learning" OR MOOC OR "online course*")
AND ("engineering education" OR "technical education" OR "computing
education"))
AND ("instructional design" OR "technology acceptance" OR "learning
design" OR "design debt" OR "knowledge taxonomy" OR "Bloom* taxonomy"))
AND PY=(2005-2025) AND LA=(English OR Spanish)
AND DT=(Article OR "Proceedings Paper" OR Review)
```

ERIC.

```
("online learning" OR "e-learning" OR MOOC OR "online course")
AND ("engineering education" OR "technical education" OR "computing
education")
AND ("instructional design" OR "technology acceptance" OR "learning
design" OR "design debt" OR "knowledge taxonomy")
```

IEEE Xplore.

```
("Abstract": "online learning" OR "Abstract": "e-learning" OR "Abstract": MOOC
OR "Abstract": "online course")
AND ("Abstract": "engineering education" OR "Abstract": "technical education"
OR "Abstract": "computing education")
AND ("Abstract": "instructional design" OR "Abstract": "technology acceptance"
OR "Abstract": "learning design" OR "Abstract": "design debt" OR "Abstract": "knowledge
taxonomy")
```

2.5 Search execution notes

2.6 Title and abstract screening criteria

Each record is judged as *yes*, *no*, or *unclear* for every criterion. A criterion marked as *mandatory* can independently justify exclusion when the judgment is *no*. A contextual criterion contributes to the holistic relevance assessment but should not independently exclude a record. An *unclear* judgment must be resolved through reviewer consensus or full-text assessment.

Table 1. Manual reproducibility package for the systematic review

ID	Stage	Required record	Verification evidence
R1	Protocol	Research questions, scope, Dated protocol and databases, period, languages, change log. eligibility criteria, reviewer roles, and conflict-resolution rule.	
R2	Search execution	Exact query used in each Search log and screen-database, search date, filters, in- shots or search-history interface, number of results, ex- export. port format, and responsible re- searcher.	
R3	Raw records	Original, unmodified exports Read-only source files from Scopus, Web of Science, with stable file names. ERIC, and IEEE Xplore.	
R4	Deduplication	Canonical record, duplicate Deduplication log and record, matching rule, reason, retained-record identi- and reviewer confirmation for fier. uncertain matches.	
R5	Title/abstract screen- ing	One row per study and reviewer, Individual reviewer judgments for the seven screen- sheets and consensus ing criteria, overall decision, ex- record. clusion reason, and brief sup- porting evidence.	
R6	Full-text screening	Final eligibility decision, page or Full-text eligibility section supporting the decision, sheet and exclusion log. and one primary exclusion rea- son for PRISMA reporting.	
R7	Snowballing	Seed study, backward or for- Snowballing log linked ward route, candidate reference, to the same screening screening decision, and contri- criteria. bution to the final corpus.	
R8	Quality appraisal	Predefined appraisal items, Completed appraisal score or judgment per item, form and reviewer reviewer, rationale, and resolu- agreement summary. tion of disagreement.	
R9	Data extraction	Structured values for the re- Extraction form and search questions, together with study-level evidence a supporting quotation and matrix. page or section. Inferred debt in- dicators are distinguished from explicit statements.	
R10	Synthesis and report- ing	Category definitions, frequency Evidence-coverage ma- tables, cross-study comparisons, trix, PRISMA checklist, PRISMA counts, and links from and final count sheet. each finding to its supporting studies.	
R11	Archiving	File inventory, version, date, au- README, mani- thor, modification description, fest, change log, and and licensing or access restric- archived release. tions. Copyrighted PDFs are not redistributed.	

Table 2. Methodological and reporting protocol

Element	Specification
Review methodology	Kitchenham and Charters (2007): planning, conducting, and reporting the systematic literature review.
Reporting work	PRISMA 2020 flow diagram and 27-item checklist.
Time span	Publications from 2005 to 2025, inclusive.
Languages	English and Spanish.
Viewpoint	Researchers and evaluators of instructional design.
Context	Higher technical education and digital learning resources.

Table 3. Goal–Question–Metric formulation

GQM nment	compo- Definition
Analyze	Online technical learning resources and their evaluation approaches.
For the purpose of	Identifying acceptance criteria and indicators of instructional design debt.
With respect to	Alignment with knowledge taxonomies, pedagogical validity, and methods for identifying instructional design debt.
From the viewpoint of	Researchers and evaluators of instructional design.
In the context of	Higher technical education and digital learning resources.

Table 4. Rationale for database selection

Database	Rationale
Scopus	Broad multidisciplinary coverage, including engineering, computing, education, and social sciences.
Web of Science	Complementary coverage of indexed journals and conference proceedings.
ERIC	Specialized coverage of education, pedagogy, and educational technology.
IEEE Xplore	Specialized coverage of engineering, computing, and technology-enhanced education.

Table 5. Database-specific execution notes

Database	Execution note
Scopus	Use Advanced Search. Execute the complete query including the W/15 proximity operator, publication years, languages, document types, and subject areas.
Web of Science	Use Advanced Search. The TS field includes Keywords Plus; this choice must be recorded because it may increase recall.
ERIC	Apply the facets <i>Publication Date: 2005–2025</i> and <i>Peer reviewed only</i> . Because an NBIB export does not independently prove that the peer-review facet was active, preserve a screenshot or search-history record.
IEEE Xplore	Use Command Search, repeating the Abstract field for each term. Apply the facets <i>Years: 2005–2025</i> and <i>Content type: Conferences and Journals</i> .

Table 6. Operational criteria for title and abstract screening

Criterion	Role	Screening question	Exclusion code
Acceptance or evaluation criteria	Contextual	Does the study discuss factors, standards, or criteria used to accept, evaluate, or validate online learning resources rather than only describing them?	EX_no_acceptance_evaluation
Knowledge taxonomies	Contextual	Does the study use or reference knowledge or competency frameworks such as Bloom, SOLO, constructive alignment, ABET, EUR-ACE, EQF, or comparable frameworks?	EX_no_knowledge_taxonomy
Online learning resource	Mandatory	Does the study examine digital materials, e-learning platforms, online courses, virtual laboratories, or web-based educational tools?	EX_not_online_learning_resource
Publication quality	Mandatory	Is the source an eligible scholarly publication, such as a journal article, conference paper, review, or verifiable institutional report?	EX_publication_quality
Research type	Mandatory	Is it empirical research or a systematic review/meta-analysis rather than an editorial, opinion paper, or commentary?	EX_wrong_research_type
Technical education context	Mandatory	Does the study examine postsecondary technical, technological, engineering, computing, STEM, vocational, or professional education, or provide evidence explicitly applicable to that context?	EX_not_technical_education_context
Instructional design deficiencies	Contextual	Does the study identify or evaluate instructional deficiencies such as misalignment, insufficient scaffolding or feedback, fragmented or outdated content, excessive cognitive load, or missing effectiveness evidence?	EX_no_instructional_design_deficiency

Primary exclusion rule. When several mandatory criteria receive a *no*, one primary PRISMA reason is assigned in the following order: (1) not an online learning resource; (2) ineligible publication quality; (3) wrong research type; and (4) outside the technical-education context. Contextual criteria may be recorded as secondary reasons, but they should not be used as the sole primary exclusion reason. Additional codes are: `EX_holistic_mismatch`, which requires a textual justification; `EX_no_fulltext`; and `EX_duplicate`.

2.7 Quality appraisal

The methodological quality of each included study is assessed using the following checklist:

Table 7. Quality-appraisal checklist

ID	Quality item
Q1	Are the study objectives clearly stated?
Q2	Is the research design appropriate for the stated objectives?
Q3	Are the data sources, participants, or analyzed materials transparently reported?
Q4	Are the findings supported by the reported evidence and analysis?
Q5	Does the study provide evidence relevant to at least one research question?

For each item, reviewers should record the judgment, a brief rationale, and the page or section supporting it. The appraisal must not be generated from study identifiers or other automatic placeholders.

2.8 Protocol quality checks

Before screening begins, the review team should confirm that:

- (i) Kitchenham is used as the review methodology and PRISMA as the reporting framework;
- (ii) both research questions are fixed before screening;
- (iii) all four databases have a documented rationale;
- (iv) the seven screening criteria include a question, role, and exclusion code;
- (v) mandatory and contextual criteria are distinguished consistently;
- (vi) the rule for assigning one primary PRISMA exclusion reason is fixed;
- (vii) the publication period and eligible languages are recorded; and
- (viii) any protocol amendment is dated and justified in the change log.

2.9 Full-Text Screening

Full-text screening applied the same seven analytical dimensions used during title and abstract screening, but with stricter evidence requirements. At this stage, a criterion was not considered satisfied solely because a term appeared in the title, keywords, or background section. Reviewers examined the complete article, including methods, results, tables, figures, and appendices when available, and recorded the page, section, table, or figure supporting each judgment.

The full-text decision comprised three gates. First, every study had to satisfy the four mandatory scope and quality criteria: it had to examine an online instructional resource, constitute an eligible scholarly publication, use an eligible research design, and address higher technical, engineering, computing, STEM, vocational, or professional education. Second, the study had to provide substantive evidence for at least one analytical relevance criterion: acceptance or evaluation criteria, operational knowledge-taxonomy alignment, or instructional-design deficiencies. Third, every positive analytical judgment had to be traceable to explicit textual or empirical evidence. A pedagogical deficiency was not coded as instructional design debt unless the article also supported persistence, accumulation, deferred remediation, recurrent consequences, or future cost.

Table 8. Operational criteria for full-text screening

Criterion	Role	Full-text verification rule	Evidence to record
Acceptance or evaluation criteria	Analytical	The study applies, validates, compares, or empirically examines criteria used to judge the acceptance, suitability, quality, or continued use of an online learning resource. A background-only mention of TAM, UTAUT, usability, or quality criteria is insufficient.	Applied construct, instrument, rubric, threshold, data source, result, and page or section.
Knowledge taxonomies	Analytical	A knowledge, cognitive, competency, or alignment framework is operationally used to design, classify, assess, or interpret the resource, its activities, or its outcomes. A superficial citation of Bloom, SOLO, ABET, EQF, EUR-ACE, or constructive alignment is insufficient.	Framework name, operational use, mapped objective/activity/assessment, and page, table, or figure.
Online learning resource	Mandatory	The unit of analysis is an instructional digital artifact, such as an online course, LMS-supported module, MOOC, virtual or remote laboratory, OER, interactive simulation, digital assessment, or web-based learning tool. Administrative or infrastructure-only systems are excluded.	Resource type, instructional purpose, delivery mode, and supporting page or section.
Publication quality	Mandatory	The complete source is an eligible scholarly publication with verifiable bibliographic information. Editorials, promotional materials, unreviewed opinion pieces, and sources whose status cannot be established are excluded.	Publication type, venue, year, DOI or persistent identifier, and peer-review status when available.
Research type	Mandatory	The study reports empirical research, a systematic or scoping review, a systematic mapping, or a sufficiently specified and evaluated framework. Purely speculative proposals without evaluation or transparent synthesis are excluded.	Study design, data or corpus, analysis method, and relevant methods section.
Technical education context	Mandatory	The evidence concerns postsecondary technical, technological, engineering, computing, STEM, vocational, or professional education, or the authors provide an explicit and defensible application to that context.	Institutional level, discipline, learner group, course or program, and page or section.
Instructional design deficiencies	Analytical	The study reports evidence of instructional misalignment, inadequate scaffolding or feedback, fragmented or outdated content, excessive cognitive load, accessibility problems with instructional consequences, weak practical or transfer support, or absent learning-effectiveness validation. An isolated deficiency must be distinguished from cumulative instructional design debt.	Deficiency, unit of analysis, explicit or inferred status, consequence, temporal or cumulative evidence, and page or section.

Full-text decision rule. A study was included when: (i) all four mandatory criteria were judged *yes*; (ii) at least one of the three analytical criteria was judged *yes*; and (iii) the supporting evidence was traceable to the full text. A judgment of *unclear* triggered independent reassessment or discussion between reviewers and had to be resolved before the final inclusion decision. The final decision set was therefore *include* or *exclude*; unresolved judgments were not carried into data extraction.

Primary exclusion reasons. Only one primary exclusion reason was assigned for PRISMA reporting, using the following order:

- (1) full text unavailable after documented retrieval attempts;
- (2) duplicate discovered during full-text inspection;
- (3) not an online instructional learning resource;
- (4) ineligible publication type or unverifiable publication quality;
- (5) ineligible research type;
- (6) outside the higher technical or professional education context;
- (7) no substantive evidence addressing acceptance, knowledge alignment, or instructional-design deficiencies; and
- (8) holistic mismatch, accompanied by a written justification.

Secondary reasons could be retained in the screening workbook, but were not added to the PRISMA totals. The absence of a knowledge taxonomy or an explicit instructional-design deficiency did not independently exclude a study when it provided substantive evidence for another analytical criterion and contributed to at least one research question.

Audit fields. For every assessed study, the full-text screening form recorded: the stable study identifier; reviewer; full-text availability; judgment for each criterion; quotation or precise paraphrase; page, section, table, or figure; overall decision; primary exclusion reason; secondary observations; date; and the outcome of any disagreement-resolution process.